

The Pasting Lemma (§1.62)

by

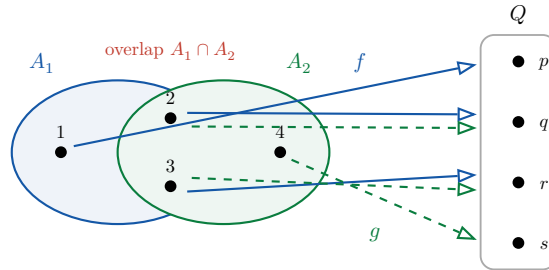
§1.62 — the intersection $\rightarrow$ union square of two subobjects is a pushout, with a concrete Set example and Freyd’s relational proof that R is a map

In a pre-logos, for subobjects $A_1, A_2 \hookrightarrow A$ with intersection $A_1 \cap A_2$ and union $A_1 \cup A_2$, this square of monics is a **pushout**:

$$\begin{array}{ccc}
 A_1 \cap A_2 & \xrightarrow{\bar{y}} & A_1 \\
 \bar{x} \downarrow & & \downarrow x \\
 A_2 & \xrightarrow{y} & A_1 \cup A_2
 \end{array}$$

Pushout means: a map $f : A_1 \rightarrow Q$ **from** A_1 and a map $g : A_2 \rightarrow Q$ **from** A_2 that **agree on the overlap** — i.e. their two restrictions to $A_1 \cap A_2$ coincide, $\bar{y}f = \bar{x}g$ (the top square commutes) — glue to a **unique** $R : A_1 \cup A_2 \rightarrow Q$ with $xR = f$ and $yR = g$.

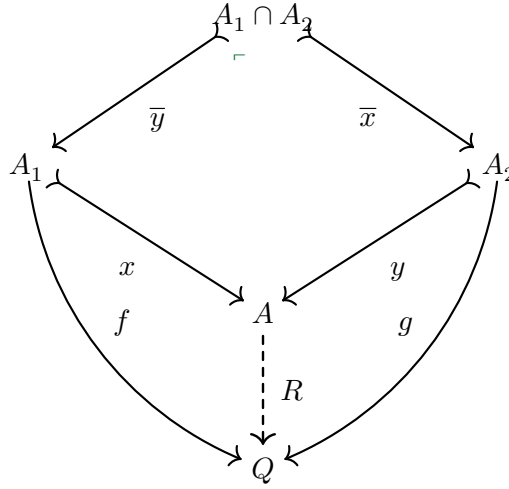
Concretely in SET: $A_1 = \{1, 2, 3\}$, $A_2 = \{2, 3, 4\}$, overlap $A_1 \cap A_2 = \{2, 3\}$, union $\{1, 2, 3, 4\}$, and $Q = \{p, q, r, s\}$. Take $f : 1 \mapsto p, 2 \mapsto q, 3 \mapsto r$ and $g : 2 \mapsto q, 3 \mapsto r, 4 \mapsto s$. The agreement is the commuting condition: on the overlap f and g **both** send $2 \mapsto q$ and $3 \mapsto r$ (the converging arrows below), so the four arrows form one well-defined $R : \{1, 2, 3, 4\} \rightarrow Q$. Had they disagreed at 2 or 3, no single R could exist — that is the whole force of “agree on the overlap”.



The pasted map $R = f \cup g$ is just **all six arrows together**; it is a well-defined function exactly because the blue (f) and green (g) arrows **coincide** on the overlap $\{2, 3\}$.

§1. The figure the book actually draws

Freyd’s proof is purely **relational** (it never names elements). The picture below is the one in the book on p.101: the pullback square on top, the two subobjects x, y landing in the union A , a test object Q with the agreeing pair f, g , and the mediator $R : A \rightarrow Q$ to be constructed. Bars name the **pullback legs** — $\bar{y}$ is the copy of y pulled back over A_1 , $\bar{x}$ the copy of x over A_2 — so $\bar{y} : A_1 \cap A_2 \rightarrow A_1$ and $\bar{x} : A_1 \cap A_2 \rightarrow A_2$.



Everything is forced by ten relational facts, exactly the list the book writes down (composition in diagram order, $R^\circ = \text{converse}$). The first eight are **generic regular-category** facts; only the cover identity needs the pre-logos.

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|--|---------------------------|---|
| (1) $xx^\circ = 1, yy^\circ = 1$ | — x, y monic | (2) $xy^\circ = \bar{y}^\circ \bar{x}$ — pullback tabulates overlap |
| (3) $1 \subseteq ff^\circ, 1 \subseteq gg^\circ$ | — f, g entire | (4) $f^\circ f \subseteq 1, g^\circ g \subseteq 1$ — f, g simple |
| (5) $\bar{x}g = \bar{y}f$ | — agree on overlap | (6) $(\bar{y}f)^\circ(\bar{y}f) \subseteq 1$ — $\bar{y}f$ simple (★) |

Cover (the one pre-logos input): $x^\circ x \cup y^\circ y = 1$. A monic z contains $\text{Im}(x), \text{Im}(y)$ iff $x^\circ x \subseteq z^\circ z$ and $y^\circ y \subseteq z^\circ z$; in a pre-logos the union $A_1 \cup A_2$ is least, so its coreflexive is 1.

Why is (6) true? The book states it without a word. Because $\bar{y}f$ is a **composite of maps**: $\bar{y} : A_1 \cap A_2 \hookrightarrow A_1$ is monic (hence a map) and $f : A_1 \rightarrow Q$ is a map, and maps are closed under composition [§1.561]. Every map is **simple** [§1.564], i.e. satisfies $h^\circ h \subseteq 1$ — so $(\bar{y}f)^\circ(\bar{y}f) \subseteq 1$. That is the **entire content** of the unexplained line; the same remark gives $f^\circ f, g^\circ g \subseteq 1$ in (4).

Why the cover $x^\circ x \cup y^\circ y = 1$? For a subobject $x : A_1 \hookrightarrow A$, the relation $x^\circ x : A \rightarrow A$ is the **coreflexive** (sub-identity) cutting out $\text{Im}(x) = A_1$: in SET, $x^\circ x = \{(a, a) : a \in A_1\}$, the diagonal on $A_1 \subseteq A$. Likewise $y^\circ y$ is the diagonal on A_2 , and their union is the diagonal on $A_1 \cup A_2$. A monic z contains both images iff $x^\circ x \subseteq z^\circ z$ **and** $y^\circ y \subseteq z^\circ z$, i.e. iff $x^\circ x \cup y^\circ y \subseteq z^\circ z$ — so $x^\circ x \cup y^\circ y$ is the coreflexive of the **least** subobject containing A_1 and A_2 , their union $A_1 \cup A_2$. The proof's object A is that union (the pushout corner, where x, y land — as in the figure above), so its coreflexive is the whole identity 1_A . It is the **pre-logos** that makes the union of subobjects exist and be computed by this relational $\cup$.

Why $xy^\circ = \bar{y}^\circ \bar{x}$ (fact (2))? It is the **tabulation** of the composite — **by construction**, not a universal-property argument. Relational composition $x; y^\circ$ is **defined** by pulling back the cospan $A_1 \xrightarrow{x} A \xleftarrow{y} A_2$; that pullback **is** the intersection $A_1 \cap A_2$, and its two projections **are** the legs $\bar{y}, \bar{x}$. Reading the relation off that jointly-monic span gives the equality $xy^\circ = \bar{y}^\circ \bar{x}$ (likewise $yx^\circ = \bar{x}^\circ \bar{y}$). Lean's `inter_lemma` records only the containment $xy^\circ \subseteq \bar{y}^\circ \bar{x}$ —

all the monotone chain needs — because it forms the intersection over the object A_1, A_2 are subobjects of rather than over their union, so it compares two pullbacks by a map and lands on $\subseteq$; over the union (the figure's A) it is the equality above.

§2. The proof that R is a map (Freyd, p.101)

Do not build a function and verify “well-defined + agrees”. Instead write down a single **relation**

$$R = x^\circ f \cup y^\circ g : A \rightarrow Q$$

and prove it is a **map**. By §1.564 a relation is a map iff it is **entire** ($1 \subseteq RR^\circ$, total) and **simple** ($R^\circ R \subseteq 1$, single-valued). Both fall out of the facts above — totality from the cover, single-valuedness from the agreement — with the only pre-logos input being the cover and §1.616 (union distributes over composition, used to multiply out the unions).

Entire. Pump each cover-branch up with the totality of f, g (fact (3)), then factor:

$$\begin{aligned} 1 &\subseteq x^\circ 1x \cup y^\circ 1y \subseteq x^\circ f f^\circ x \cup y^\circ g g^\circ y \\ &\subseteq (x^\circ f \cup y^\circ g)(f^\circ x \cup g^\circ y) \subseteq RR^\circ. \end{aligned}$$

Simple. Multiply out (§1.616), kill the diagonal monic loops by (1), reroute the cross terms by the tabulation (2), and fold them together by the agreement (5):

$$\begin{aligned} R^\circ R &\subseteq (f^\circ x \cup g^\circ y)(x^\circ f \cup y^\circ g) \\ &\subseteq f^\circ x x^\circ f \cup f^\circ x y^\circ g \cup g^\circ y x^\circ f \cup g^\circ y y^\circ g \\ &\subseteq f^\circ f \cup f^\circ \bar{y}^\circ \bar{x} g \cup g^\circ \bar{x}^\circ \bar{y} f \cup g^\circ g \\ &\subseteq 1 \cup (\bar{y} f)^\circ \bar{x} g \cup (\bar{x} g)^\circ \bar{y} f \cup 1 \\ &\subseteq 1 \cup (\bar{y} f)^\circ \bar{y} f \cup (\bar{y} f)^\circ \bar{y} f \subseteq 1. \end{aligned}$$

Since $1 \subseteq RR^\circ$ and $R^\circ R \subseteq 1$, **R is a map** [§1.564]. It is the mediator: by (1), (2), (5),

$$xR = x(x^\circ f \cup y^\circ g) = x x^\circ f \cup x y^\circ g = f \cup \bar{y}^\circ \bar{x} g = f \cup \bar{y}^\circ \bar{y} f = (1 \cup \bar{y}^\circ \bar{y})f = f,$$

and symmetrically $yR = g$. Uniqueness of R is the cover once more: x, y are jointly epic.

★ Punchline

The mediating map is forced by the relational calculus: $R = x^\circ f \cup y^\circ g$ is **entire** because the cover splits 1_A and the totality of f, g pushes each branch into RR° ; **simple** because $R^\circ R$ multiplies out into four terms that each reduce to $\subseteq 1_Q$ (the diagonals by the monic $xx^\circ = 1$ and the map-cap $f^\circ f \subseteq 1$; the cross terms by the tabulation $xy^\circ = \bar{y}^\circ \bar{x}$, the agreement (5), and the map-cap of $\bar{y}f$); and **unique** because x, y are jointly epic. The one step that needs the pre-logos is $\cup$ distributing over composition (§1.616) — which is why the lemma fails additively (1.622: it breaks for groups).

§3. §1.622 — when pasting can fail

The §1.62 proof has one non-formal input: $\cup$ distributes over composition (§1.616). Where that fails, so can the lemma. §1.622 records two faces of this.

Abelian categories — still a pushout, but for a different reason. There $A_1 \cup A_2 = A_1 + A_2$, and the intersection $\rightarrow$ union square is **bicartesian** (pullback = pushout) straight from the additive structure (the second isomorphism theorem). Freyd’s relational proof does **not** apply: unions of relations do not distribute over composition in any non-degenerate additive regular category.

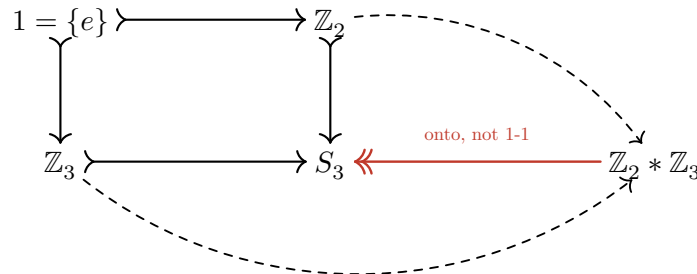
Groups — not a pushout. S_3 is the group of all **six** rearrangements of three corners 0, 1, 2 (the symmetries of a triangle): three rotations 012 (do nothing), 120, 201, and three swaps 102, 021, 210 (each list records where corners 0, 1, 2 go). Inside it sit two subgroups:

- $A_1 = \mathbb{Z}_2 = \{012, s\}$, the identity and the swap $s = 102$ — order 2, since s twice is the identity;
- $A_2 = \mathbb{Z}_3 = \{012, r, r^2\}$, the identity and the rotation $r = 120$ — order 3.

The meet $A_1 \cap A_2 = 1$. An intersection of subgroups is a subgroup of each, so by Lagrange its size divides both 2 and 3 — the only common divisor is 1, leaving just the identity 012 (the one list in both). Coprime orders force a trivial overlap.

The union $A_1 \cup A_2 = S_3$ — and this is the crux. The lattice “union” is **not** set-union: $\{012, 102, 120, 201\}$ is not even a subgroup, since composing s with r gives 021, outside the list. The **join** is the **smallest subgroup containing both** — the subgroup **generated** by s and r (close that set under composition). A swap and a rotation generate everything, so $\langle s, r \rangle = S_3$ (its order is divisible by 2 and 3, hence by $6 = |S_3|$).

This closing-under-composition is exactly where groups depart from SET: there the join of two **subsets** simply **is** their set-union, with nothing to close — so pasting always works. §1.62 would make the square below a pushout — it is not.



The true pushout of $\mathbb{Z}_2 \leftarrow 1 \rightarrow \mathbb{Z}_3$ in GRP is the **free product** $\mathbb{Z}_2 * \mathbb{Z}_3$ — the **infinite** modular group $\text{PSL}_2(\mathbb{Z})$. The comparison $\mathbb{Z}_2 * \mathbb{Z}_3 \twoheadrightarrow S_3$ is onto but far from injective: S_3 is the proper quotient that imposes the dihedral relation $srs = r^2$ (a swap conjugates the rotation to its inverse), which the free product never sees. A finite group is not the gluing; the genuine gluing is infinite.

Pasting literally fails here. Put $Q = \mathbb{Z}_2 * \mathbb{Z}_3$ and let $f : \mathbb{Z}_2 \rightarrow Q$ and $g : \mathbb{Z}_3 \rightarrow Q$ be the two inclusions. They **agree on the overlap** 1 (both fix e) — yet **no** homomorphism $R : S_3 \rightarrow Q$ extends them: its image would be a **finite** subgroup of Q containing both factors, but those two already generate the **infinite** Q . Two maps agreeing on the overlap with nothing to glue them onto: exactly the failure of “paste”. In SET (or any pre-logos) this cannot happen — §1.616 guarantees the gluing exists and is the union.